

**The ADEQUATe platform: data- and  
community-driven quality improvements**

# Search, Filter, Fork, and Link Open Data



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*The project is based on two observations...*

- An increasing amount of Open Data becomes available
- The integration of such open, freely re-usable data sources is a key success factor

*... and two identified issues:*

- The quality issues with metadata and the data itself
- The lack of interoperability between data sources

✓ **2 Austrian Use Case Portals:** [data.gv.at](http://data.gv.at) & [opendataportal.at](http://opendataportal.at)

## ✓ *Four focus group interviews, addressing different stakeholder*

- Lay-persons @ Vienna Open Data meet-up
  - Open (Government) Data interested @ GovCamp Vienna
  - Platform providers @ Austrian OGD Platform meeting
  - Scientists @ IRIS conference 2016
- 106 attendees

## ✓ *Online based survey*

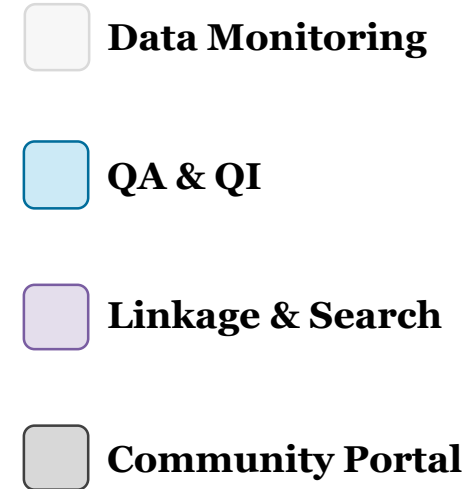
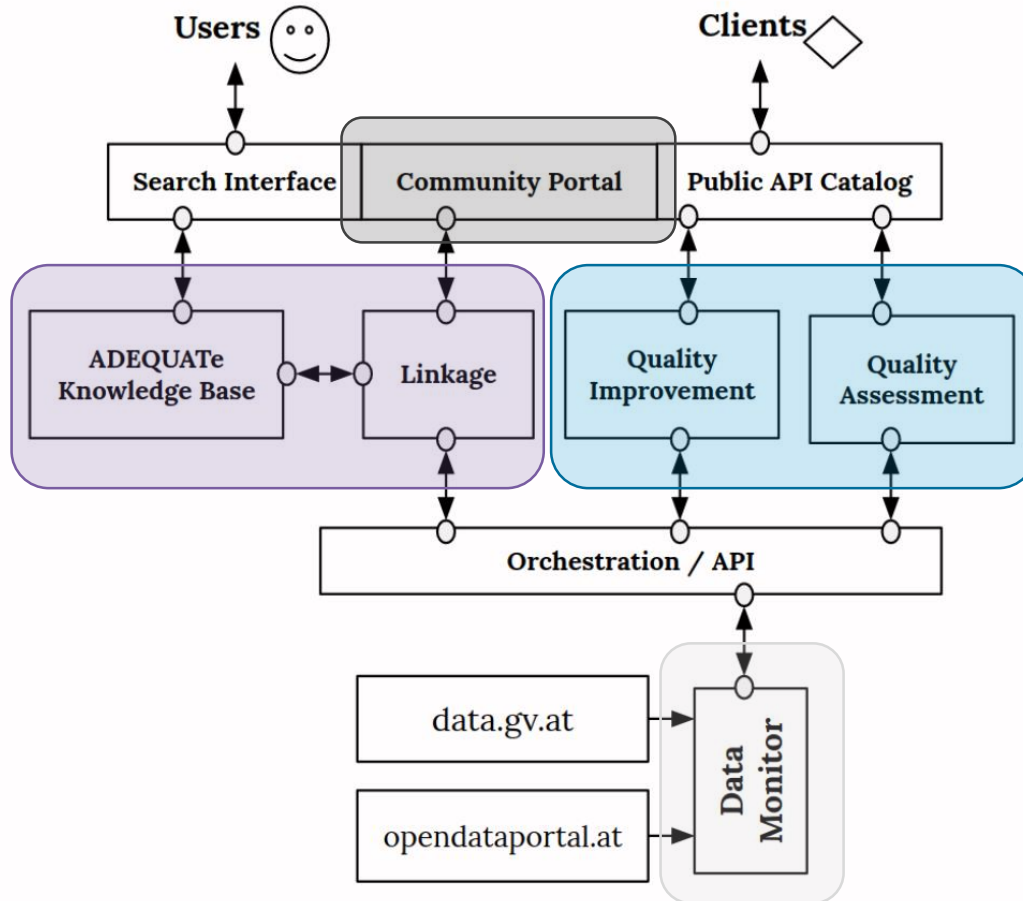
- Data quality issues on Austrian data portals
  - Sent to various channels: community mailing lists, social media groups, etc.
- 183 participants

## *Users want..<sup>1</sup>*

- Consistent data formats, complete and correct descriptions, up-to-date data, clarity about legal aspects **Quality report**
- See issues, changes and development of datasets over time **Quality improvements**
- Find all resources for a certain entity (e.g. city), filter based on concepts **Linkage & Search**
- Discuss and collaborate on datasets, fork/work on own versions **Collaboration**

<sup>1</sup> We already filtered out requirements that are beyond our control, e.g., that users want the public data to be qualified as correct.

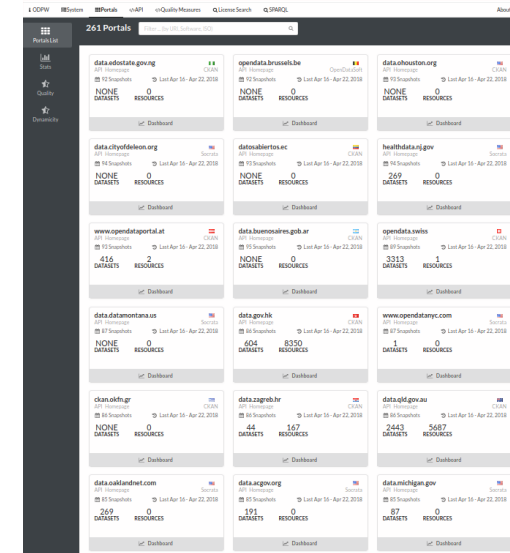
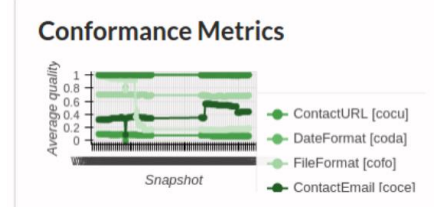
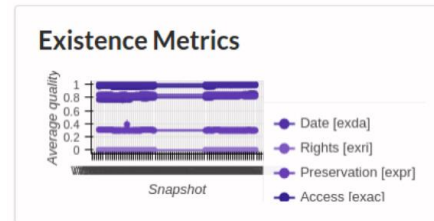
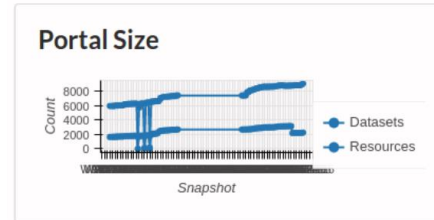
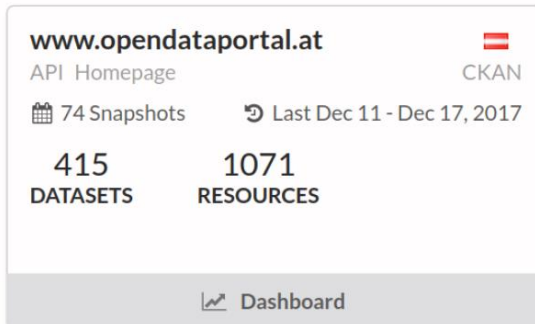
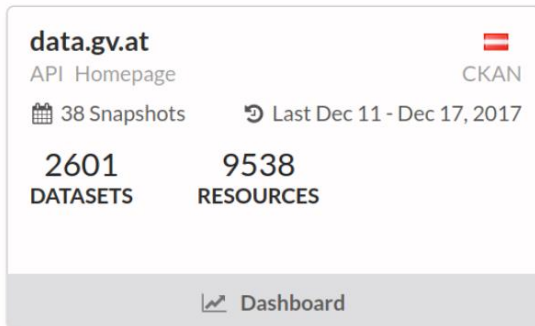
# The Platform



# (i) Data Monitoring

**Monitors** and **reports** the quality of the use case partner portals

Based on **Open Data Portal Watch**  
Monitoring and QA of > 250 data portals



<http://data.wu.ac.at/portalwatch>

## (ii) Quality Assessment ...

Does the meta data contain information to contact the data provider or publisher?

Contact **Yes**

Is the used license conform to the open definition?

License Openness **Yes**

Does the meta data contain information about format, size or update frequency of the resources?

Preservation **0.50**

*Information (size, format, mimetype,...) for preserving/archiving the dataset resource are missing*

Are the available values of contact properties valid HTTP URLs?

ContactURL **No**

*The publisher or contact URL is not a syntactically valid URI*

Is the file format based on an open standard?

Format Openness **Yes**

Does the meta data contain access information for the resources?

Access **Yes**

Are the available values of access properties valid HTTP URLs?

AccessURL **Yes**

Does the meta data contain information about the license of the dataset or resource?

Rights **Yes**



## (ii) ... & Quality Improvements

### Available Services

#### CSV Clean

Parse & clean

A service to parse and clean various types of CSV derivations (or character-separated-values files). The cleaned file is UTF-8 encoded and uses the "," as value separator.

#### CSV Profiler

Metrics & statistics

This service analyses the input CSV and provides basic informations and metrics such as the encoding, column data types, and the completeness of columns.

#### CSV MetaData Editor

Describe & enrich

This is a prototypical form for generating metadata about a submitted CSV file. The metadata is compliant with the CSV on the Web metadata specification.



```
"dialect": {  
  "delimiter": ",",  
  "doubleQuote": true,  
  "encoding": "utf-8",  
  "header": true,  
},  
"tableSchema": {  
  "columns": [  
    {  
      "datatype": "anyURI",  
      "name": "artist_id"  
    },  
    {  
      "datatype": "anyURI",  
      "name": "name"  
    },  
    ...  
  ]  
}
```

[www.adequate.at](http://www.adequate.at)

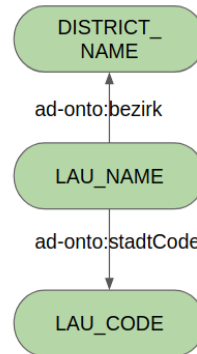


# (iii) Linkage ...

## Odalic [1]:

- Semantic interpretations of tabular data
- Annotations of (i) columns, (ii) cell values, and (iii) relations between columns
- Re-publishing as RDF

| DISTRICT_NAME           | LAU_CODE | LAU_NAME              |
|-------------------------|----------|-----------------------|
| Bezirk (ad-onto:Bezirk) |          | Stadt (ad-onto:Stadt) |
| Graz-Stadt              | 60101    | Graz                  |
| Deutschlandsberg        | 60301    | Aibl                  |
| Deutschlandsberg        | 60302    | Deutschlandsberg      |



ENRICHED CSV

RDF (TURTLE)

- Using ADEQUATe knowledge base (Austrian-specific knowledge)

**ADEQUATe Knowledge Base**

- Austrian States and Cities (9)
  - Burgenland (1)
  - Kärnten (136)
  - Niederösterreich (221)
  - Oberösterreich (434)
  - Salzburg (0)
  - Steiermark (539)
    - Admont (0)
    - Aflenz Kurort (0)
    - Aflenz Land (0)
    - Aibl (0)

**Aibl**

<https://adequate-project.pp>

Ort , Stadt

**Details** Notes

**SKOS** ADE

**Broader Concepts**

[Steiermark](#)

## (iii) ... & Search

### *PoolParty GraphSearch:*<sup>1</sup>

- Full text search
- Faceted search, based on entities in ADEQUATe knowledge base
- *TODO*: Integrating Odalic annotations in the GraphSearch

► Austrian States and Cities

► Data Publishers

► File Formats

▼ Licenses

☒ Creative Commons Namensnennung 3.0 Österreich (550)

► Tags

► Data annotations

Reset

Search...

Wien x Creative Commons Namensnennung 3.0 Österreich x

550 results

01/11/2018 | Adequate Open Data

**Nationalratswahlen Wien (NR17-wahlkreisverband)**

Ergebnisse der Nationalratswahlen in Wien.

Tags: Creative Commons Namensnennung 3.0 Österreich, CSV, Stadt Wien, Wien

► more like this

<sup>1</sup> <https://www.poolparty.biz/poolparty-semantic-graph-search-server/>

# Exemplary User Journey

Suchbegriff (z.B. Finanzen, Wahlen)

Suche starten

Datenkatalog Apps &amp; News

→ Katalog durchstöbern

Startseite

Daten ▾

Dokumente ▾

Anwendungen ▾

Infos ▾

## Katalog

### Hunde pro Bezirk Wien

Anzahl der Hunde pro Bezirk und Rasse

Daten und Ressourcen



Anzahl der Hunde pro Bezirk

Entdecke ▾

Titel und Beschreibung Englisch ⓘ

Number of dogs per district and race.

Veröffentlichende Stelle ⓘ

Stadt Wien

Kontaktseite der veröffentlichenden Stelle ⓘ

<https://open.wien.gv.at>

Veröffentlichende Stelle - E-Mailkontakt ⓘ

[open@post.wien.gv.at](mailto:open@post.wien.gv.at)

Datenverantwortliche Stelle ⓘ

Magistratsabteilung 6 - Rechnungs- und Abgabenwesen

ADEQUATE Checked ✓

Veröffentlichende Organisation  
bzw. Person ⓘ

Stadt Wien

Kategorie ⓘ

Verwaltung und Politik

Schlagworte ⓘ

hunde

statistik

13

# Hunde pro Bezirk Wien

The information below and the linked dataset(s) are provided by the [ADEQUATe Open Data project](#). The information and data were harvested from data.gv.at, analysed and may have been automatically and manually changed.

## Description

Anzahl der Hunde pro Bezirk und Rasse

### Published

  12/15/2015

  1/5/2018

 Stadt Wien

 Magistratsabteilung 6 - Rechnungs- und Abgabewesen

  [post@ma06.wien.gv.at](mailto:post@ma06.wien.gv.at)

 [Status of the maintainer's datasets](#) 

### Keywords

#statistik #hunde

## Downloads



Git



Metadata

### Anzahl der Hunde pro Bezirk

 CSV ·  369.2 kB ·  cc-by-at-30 ·  12/15/2015 ·  DefaultDelimiter, DropCommentLines

Original from [www.wien.gv.at](http://www.wien.gv.at): [Anzahl der Hunde pro Bezirk](#) 



Semantic linking

Files (522 KB)

Commits (29)

Branch (1)

Tags (0)

Readme

master ▾

stadt-wien\_anzahlderhundeprobezirkderstadtwien

History

🔍 Find file

 ▾

/ 

+



1815

Portal Monitor GitBot committed 6 days ago

c0100dcc



| Name                                                                                              | Last commit | Last Update  |
|---------------------------------------------------------------------------------------------------|-------------|--------------|
|  cleaned         | 1750        | 4 months ago |
|  resources       | 1750        | 4 months ago |
|  README.md       | 1811        | a month ago  |
|  metadata.jsonld | 1815        | 6 days ago   |
|  original.json | 1802        | 3 months ago |

# Outlook & Open Challenge

Platform, source code, and data online available:

- [www.adequate.at](http://www.adequate.at)

## *Outlook:*

- Integration of other QI tools, e.g. Open Refine
- Continuation in substantively related Austrian project CommuniData:  
[www.communidata.at](http://www.communidata.at)

## *Open Challenge:*

- We have the tools, services & platform...  
**How to get the users?**



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**Thank you!**

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